

Instructions: Choose the best answer for each question. Take $g = 10 \text{ m/s}^2$ where applicable.

1. For a compound pendulum obeying the small angle approximation, which of the following differential equations correctly describes its motion?

- (a) $\theta'' + \frac{g}{L}\theta = 0$
- (b) $\theta'' + \frac{Mga}{I}\theta = 0$
- (c) $\theta'' + \frac{I}{Mga}\theta = 0$
- (d) $\theta'' + \frac{C}{I}\theta = 0$

2. In the period formula for a compound pendulum, $T = 2\pi\sqrt{\frac{I}{Mga}}$, what does the variable a specifically represent?

- (a) The total length of the pendulum rod.
- (b) The distance from the pivot point to the bottom of the pendulum.
- (c) The distance from the pivot point to the center of mass G .
- (d) The amplitude of oscillation.

3. A torsion pendulum consists of a disk suspended by a wire. If you replace the wire with one that has a torsion constant C four times larger ($4C$), how does the new period T' compare to the original period T ?

- (a) $T' = 2T$
- (b) $T' = T$
- (c) $T' = \frac{T}{2}$
- (d) $T' = \frac{T}{4}$

4. Why is the condition $\theta_m \leq 10^\circ$ necessary for the simple harmonic motion model of a simple pendulum?

- (a) To ensure air resistance is negligible.
- (b) To ensure the restoring force is linear (Linearization).
- (c) To prevent the string from breaking.
- (d) To keep the mechanical energy conserved.

5. A simple pendulum has a mass m and length L . If the mass of the bob is doubled to $2m$, what happens to its proper period T_0 ?

- (a) It increases by a factor of $\sqrt{2}$.
- (b) It remains unchanged.

- (c) It decreases by a factor of $\sqrt{2}$.
- (d) It doubles.
6. In a Torsion Pendulum, which energy transformation occurs as the disk moves from its maximum angular displacement $+\theta_m$ to the equilibrium position?
- (a) Kinetic Energy converts to Gravitational Potential Energy.
- (b) Elastic Potential Energy converts to Kinetic Energy.
- (c) Kinetic Energy converts to Elastic Potential Energy.
- (d) Gravitational Potential Energy converts to Kinetic Energy.
7. You are analyzing a compound pendulum rod of mass M and length L . The pivot is at one end. To calculate the period, you need the Moment of Inertia I . Which approach is correct?
- (a) Use $I = \frac{1}{12}ML^2$ directly.
- (b) Use the Parallel Axis Theorem: $I = \frac{1}{12}ML^2 + M(\frac{L}{2})^2$.
- (c) Use $I = ML^2$ directly.
- (d) Assume I is negligible.
8. If the mechanical energy of a free torsion pendulum decreases over time, what can you conclude about the system?
- (a) The torsion constant C is increasing.
- (b) The motion is damped by non-conservative forces.
- (c) The system is being driven by an external force.
- (d) The small angle approximation has been violated.
9. Which of the following physical quantities is the rotational analog to the spring constant k in a linear spring-mass system?
- (a) Moment of Inertia (I)
- (b) Torsion Constant (C)
- (c) Gravitational Acceleration (g)
- (d) Angular Amplitude (θ_m)
10. For a compound pendulum, if the pivot point is moved closer to the center of mass ($a \rightarrow 0$), what happens to the period T ?
- (a) The period approaches zero.
- (b) The period approaches infinity.
- (c) The period remains constant.
- (d) The period decreases linearly.